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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	NoMoreTokens
Project / Solution Name	TomorrowWorks
Name (Leader)	Lee Yun Loong
Email & Phone (Leader)	yunloong6325@gmail.com / +6018-324 8843
University Affiliation(s)	Multimedia University (MMU), Cyberjaya
Live Prototype URL	https://tomorrow-works.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/kairos8232/TomorrowWorks

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Browser: a React 19 + TypeScript single-page app (built with Vite) runs the scoring rule, SDG rules, futures simulator, learning plan and CV checks on committed JSON snapshots (DOSM, ILO, ESCO, UN SDG, hand-checked course and NOSS records).

Server: a dependency-free Node.js 20 server on Replit Autoscale serves the built site and the routes /api/skill-search, /api/cv-skills, /api/news, /api/lfs, /api/dosm-next and /api/health.

AI: the server sends the person's text to a RunPod Serverless endpoint (Infinity embedding worker, BAAI/bge-m3), then ranks the 13,960 ESCO skill vectors it holds in memory; those vectors were made once on a RunPod GPU Pod (Job A). The nearest real ESCO skills return to the browser, and the person ticks the true ones.

Live data: the server fetches GDELT news and OpenDOSM labour-force figures, and checks DOSM's release page for a newer Job Market Insights quarter, at most once a day (memory only; it never imports data itself) and falls back to the committed snapshots. There is no database and no user data is stored. The RunPod key and endpoint ID are Replit Secrets, read only on the server. If RunPod is unavailable, a searchable ESCO skill list is used.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Language for the whole front end and tests	Apache-2.0
React 19.2.7 and react-dom 19.2.7	User interface	MIT
Vite 6.4.3	Build tool and dev server	MIT
Vitest 5.0.5	Automated tests (140)	MIT

Name & Version	Purpose in Project	License
pdfjs-dist 5.7.284	Reads PDF CVs in the browser (nothing uploaded as a file)	Apache-2.0
@fontsource/carlito 5.3.0	Web font (skipped in Low-data mode)	OFL-1.1
Node.js 20 or newer (no server packages)	Server runtime	MIT
Python 3.12: sentence-transformers, torch 2.8.0 (RunPod Pod); NumPy, openpyxl (importer scripts on our computers)	Embedding job and data importers; not part of the running app	Apache-2.0; BSD-3-Clause; BSD-3-Clause; MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless endpoint (Infinity embedding worker, OpenAI-compatible /embeddings route), RunPod	Embeds the person's text for ESCO skill search; also used in batch to match courses and UN SDG targets	Sponsor-provided credit (US$25)
GDELT DOC 2.0 API, The GDELT Project	Malaysian news headline metadata for a context page; never affects scores	Free, no key
DOSM release page (dosm.gov.my), DOSM	Once a day, checks whether a newer Job Market Insights quarter is published; scores change only after a person imports it	Free, no key
OpenDOSM data catalogue (lfs_month), DOSM	Monthly labour-force figures shown as context	Free, open data
SDG API, UN Statistics Division	Official text of the numbered SDG targets (one-off import)	Free
Coursera public course catalogue (courses.v1), Coursera	Course names, links and stated workload for AI-matched suggestions (one-off import)	Free public endpoint
Hugging Face dataset download, Anthropic Economic Index	One-off import of AI-use figures	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Autoscale deployment, Secrets, Shell)	Hosts the app and the Node server; holds the RunPod key and endpoint ID; Shell runs one importer	Sponsor-provided credit (US$20); Autoscale, max 1 machine; region
RunPod GPU Pod (Job A), RunPod	Embedded all 13,960 ESCO skills once: Pod rear_amethyst_gull (ID 40jrr8aulh76tb), NVIDIA RTX A5000, 92.5 s; Pod stopped afterwards	On-demand Pod; data centre EU-SE-1
RunPod Serverless endpoint j6ssjfy9z074hp (Job B), RunPod	Serves bge-m3 embeddings for skill search and batch matching	16 GB GPU pool; Active workers 1, idle 5 s; data centres not restricted (RunPod picks any available one)
GitHub	Public source repository (kairos8232/TomorrowWorks), Apache-2.0	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
BAAI/bge-m3 (via RunPod Infinity Embedding worker 1.1.4)	BAAI, open weights on Hugging Face	Pre-trained, used as is (no fine-tuning); turns text into vectors to find existing ESCO skills, courses and SDG targets	MIT

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
DOSM Big Data Analytics: Job Market Insights, Q1 and Q2 2026	https://www.dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-q22026	Public	DOSM terms of use, cited	N
ILO Working Paper 140, Table A1 (GenAI exposure by occupation), 2025	https://webapps.ilo.org/static/english/intserv/working-papers/wp140/index.html	Public	ILO publication terms, cited	N
ESCO v1.2.1 classification (skills, occupations); our 9-role list is an adapted selection	https://esco.ec.europa.eu	Public	Free reuse with attribution (Commission Decision 2011/833/EU)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM Monthly Principal Labour Force Statistics (lfs_month)	https://open.dosm.gov.my/data-catalogue/lfs_month	Public	CC BY 4.0	N
GDELT DOC 2.0 news metadata (Malaysian outlets)	https://www.gdeltproject.org/	Public	GDELT terms (about page), cited	N
Microsoft Research, Working with AI, v1.1	https://github.com/microsoft/working-with-ai	Public	CC BY 4.0	N
Anthropic Economic Index	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC-BY (data)	N
UN SDG numbered target list (UNSD SDG API)	https://unstats.un.org/SDGAPI/v1/sdg/Target/List	Public	UN terms, cited	N
Coursera course catalogue (name, link, stated workload only)	Coursera courses.v1 public endpoint	Public	Coursera terms; no course content copied	N
Hand-checked course records: OpenLearn, CS50x, UN CC:Learn; Rakyat Digital access note	Provider pages (links in data/pathway/resources.json)	Public	Provider terms; links and stated hours only	N
Career-advice articles behind the CV checks (Jobstreet Malaysia and Singapore, Jobscan, Santa Clara University Career Center)	Links in src/cvChecks.ts	Public (web articles; our rules, no text copied)	Each publisher's terms; cited	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
NOSS / SKM lists: JPK Daftar Standard (22 Jan 2025), JPA SKM list (1 Dec 2024)	myspike.my and docs.jpa.gov.my (links in data/pathway/noss.json)	Public	Government publications, cited	N
Work-interest profiles: job tag (Japan MHLW) and O*NET OnLine	https://shigoto.mhlw.go.jp ; https://www.onetcenter.org	Public	job tag terms; O*NET CC BY 4.0	N
SDG review sheet: 145 skill-target pairs labelled by two team reviewers	data/sdg/sdg-review.json in our repository	Self-collected	Apache-2.0 (our work)	N
Sample people, teams and CV text (Aina, Hakim, Puan Rosnah, Encik Lim, Encik Faizal)	src/data.ts, src/sampleCv.ts in our repository	Synthetic	Apache-2.0 (our work)	N: all fictional

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Anthropic Claude Opus 5.5 and Sonnet 5.5	Writing and reviewing code, importers, tests and documentation	Major
OpenAI Codex	Early planning, project documents, RunPod tool set-up and design passes	Moderate
Replit Agent	Automatic save and publish commits when the app was published; no feature code	Minor
Impeccable design skill	Design review suggestions during the UI pass	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every change is committed in small steps with its checklist update, and 140 automated tests (scoring rules, worked example, SDG rules, news rules, fairness checks) must pass. Our rule is that the team must be able to explain every line; Each of the members reviews and approves each change. AI never creates data: figures are imported by scripts from the original publishers, and each file carries source, URL and dates. AI-matched courses are labelled as not hand-checked, and the SDG links used in ranking come from our two-reviewer blind check.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod (SDC sponsor)	US$25 credit on the team's RunPod account	Paid for the GPU Pod that embedded the ESCO skills and the Serverless endpoint that serves skill search and the course and SDG matching
Replit (SDC sponsor)	US$20 credit on the team's account	Hosts the live app and its Node server, keeps the RunPod key and endpoint ID as Secrets, and runs one data import in the Shell

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit hosts the app, and the RunPod GPU work powers skill search and the AI matching; the handbook requires both, but the app does not depend on a live RunPod call. The 13,960 skill vectors are saved in the repository, so with RunPod down the app still works from a searchable ESCO list. Another embedding provider could replace the endpoint because it only turns text into vectors; the first search after idle can take about a minute while the worker starts.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No account, no database: nothing about a person is saved on our servers.

A person may type a description of their work, or paste or upload a CV or a job ad. PDF and text are read in the browser; contact lines are removed and common name, age and identity phrases are stripped there. Only the remaining text (up to 500 characters per search, or up to 40 lines of 300 characters for a CV or ad) goes to our Replit server, which passes it to RunPod to be embedded and does not store or log it; the visitor's address is held in memory only to limit searches to 20 per minute. The CV checks run in the browser only. The skills profile disappears on refresh; only language and Low-data mode are kept in localStorage. Sample people are fictional. The RunPod key is in Replit Secrets.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Search quality: embeddings can suggest a wrong skill, so the person confirms every skill. ESCO has no Malay or Chinese labels and the Commission does not guarantee its accuracy.

CV intake: scanned PDFs without a text layer cannot be read, and contact-line removal is rule-based, so unusual layouts can slip through; Check my CV uses fixed, cited rules, not a full review.

Bias: our audit found ethnicity words pulled in language skills (63 suggestions across 12 descriptions, against 0 for neutral phrases); we strip common identity phrases first, but some phrasings can still pass, so this is reduced, not solved.

Matching: AI-matched courses are labelled unchecked; AI alone found only 21 relevant SDG links out of 73, so only team-reviewed links are used.

Data: DOSM gives demand by occupation group only; scores are estimates, not forecasts.

Dependency: one embedding provider; GDELT can rate-limit, so saved snapshots are the fallback.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24/10/26